

Simons Dissertation Fellowship Proposal

My research revolves around *derived categories* in algebraic geometry. A derived category is built from (and in some sense improves) a collection of algebraic objects satisfying certain axioms, called an abelian category (for example, the collection of all vector spaces over a given field or abelian groups). In algebraic geometry, we consider the abelian category of coherent sheaves on a variety/scheme/space X , where an individual coherent sheaf is a way to attach algebraically varying families of vector spaces to X .

The derived category of (coherent sheaves of) X can be considered a more flexible version of the category of vector bundles/coherent sheaves on X , but it should also be thought of as a “higher invariant” [10], capturing data such as Hochschild cohomology, K-theory, and some Hodge theory. My reasons for interest in the derived category are the following:

- The existence of *semiorthogonal decompositions*, which describe how one derived category is assembled from smaller pieces.
- The existence of interesting derived equivalences between varieties, as in the study of *Fourier-Mukai transforms*, and derived equivalences between varieties and noncommutative algebras (like quiver algebras), as in the study of *tilting theory*.
- The objects in a derived category having good moduli theory—in other words, we can parameterize certain objects of the derived category using a space. More precisely, we parameterize *Bridgeland stable* objects, and our moduli/parameter space changes as we vary a *stability condition* to cross a wall in a certain complex manifold.

In more detail, equipping a derived category with a Bridgeland stability condition usually involves first finding a coarser object, called a t-structure. A stability condition and t-structure come in an “algebraic” flavor, when the t-structure is transferred over from a quiver via tilting theory, as in [7, 12], and then Bridgeland moduli coincide with GIT quiver moduli [11], as used by [2]. The “geometric” flavor of Bridgeland stability comes about by crafting t-structures from torsion pairs using the Bogomolov inequality, as carried out in Arcara-Bertram for all smooth projective surfaces [3].

As far as the immediate future, I am currently working on a project with my advisor Aaron Bertram, and my peers Jonathon Fleck and Liebo Pan, about applying Bridgeland stability to find further extensions of Reider’s theorem and (for surfaces) generalizations of certain theorems controlling the syzygies of a curve embedded by an adjoint divisor.

Research Plan.

With Bertram et al, we are seeking to answer questions of the following form, where S is a surface, D is a divisor, and $|K_S + D|^\vee$ the target of the morphism induced by the divisor $K_S + D$, which is the projective space of hyperplanes in $H^0(S, \omega_S(D))$.

Question. *What numerical condition do we need on D to guarantee a given property of $S \dashrightarrow |K_S + D|$?*

If we replace the surface S with a curve C for a moment, the following theorems are known (the first being an almost immediate corollary of Riemann-Roch, the last from [14]).

Numerical Condition	Property of Map
$\deg D > d + 1$	$K_C + D$ is d -very ample (i.e., separates $(d + 1)$ -jets).
$\deg D > 2$	The homogeneous ideal I_C of $C \subseteq K_C + D ^\vee$ is generated by cubics.
$\deg D > 3$	The homogeneous ideal I_C of $C \subseteq K_C + D ^\vee$ is generated by quadrics.

Continuing, one can then ask not just about generation of I_C , but also about higher syzygy modules in the minimal resolution of I_C . On the other hand, for surfaces, the starting point is Reid's theorem, which gives the following:

Numerical Condition	Property of Map
$D^2 > (d+1)^2$ and $D \cdot C > C^2 + d$ for all curves with $C^2 \leq 0$	$H^1(S, \omega_S(D) \otimes I_Z) = 0$ for all dimension 0 subschemes of length $\leq d$ (in particular $K_S + D$ is $(d - 1)$ -very ample)

This serves as a version of Riemann-Roch for surfaces, and the numerical conditions are sharp. We would then like a theorem controlling generators of I_S , and eventually higher syzygies. However, instead of directly asking for I_S to be generated by degree d polynomials, we prove a weaker condition, which is the nefness/positivity of $dH - E$ on the blowup $\text{bl}_S|K_S + D|^\vee$, where H is the hyperplane class and E is the exceptional divisor.

To prove this, we employ Bridgeland stability and a strategy of [1], where we reinterpret the ambient projective space $|K_S + D|^\vee$ as a family of Bridgeland stable complexes on S , and in turn $\text{bl}_S|K_S + D|^\vee$ is a family created by wall-crossing on the stability manifold, provided we have our numerical conditions to control the position of walls. We can then control the positivity of $dH - E$ by relating it to the Bayer-Macri nef divisor class which varies linearly in chambers of the stability manifold [4].

Career Goals.

Beyond this current research project, I have aspirations to produce more semiorthogonal decompositions via birational geometry and representation theory. On the horizon, I am excited by the ideas of [16, 15] producing semiorthogonal decompositions for moduli spaces of vector bundles on a curve, consisting of derived categories of symmetric powers of the curve. I like two variations of this direction:

- For a genus g hyperelliptic curve associated to a quadric, a theorem of Ramanan [8] describes such moduli spaces as Fano varieties of $g - 2$ dimensional linear subspaces, so there is good reason [5] to conjecture similar semiorthogonal decompositions for Fano varieties of linear subspaces of different dimensions.
- Following the analogy associating quivers with curves and moduli of quiver representations with moduli of vector bundles [9], one can also ask for semiorthogonal decompositions of moduli of quiver representations using derived categories of quivers [13, 6].

Professionally, my plan is to continue in academia, as I am very much enjoying my experience as a PhD candidate, with respect to research, organizing seminars, and being an instructor. I am enthused by my math, and I want to share it with others.

References

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